U1

- [ ] Derive Gibbs Phase Rule (NH_4Cl example ★)
- [ ] Water System (All States + Theory)
- [ ] M.O.I derivation
- [ ] Franck-Condon Principle
- [ ] Pb-Ag System (& Significance of Eutectic Point & Pattinson's Process)
- [ ] Energy Level diagram (Refer extra ques)
- [ ] Numericals (Refer from extra ques)
- [ ] Define Phase, Component, D.O.F
- [ ] Rotational Spectra (Theory + Examples) & Vibrational
- [ ] Born-Oppenheimer Approximation

U2

- ☐ Cell construction Numericals (Refer Extra Ques) & Nernst Equation ☆☆☆
- ☐ Numericals subjective to Zn-Hg, Calomel
- ☐ Ion Selective Electrodes → Glass Electrode (Theory) ☆☆☆
  → Alkaline Error
- ☐ Corrosion Complete

(Anything can be asked in complete random order)

V3

- [ ] E.D, P.D, ESD Numericals
- [ ] Battery & Characteristics
- [ ] Zn-Air Battery
- [ ] H_2O_2 Fuel Cell
- [ ] Oxygen Sensor
- [ ] Ragone Plot ★★
- [ ] Li-Ion Battery (Advantage over Li, All theory ★)
- [ ] Reserve Batteries
- [ ] Super Capacitors
- [ ] Mg - Water Activated Battery
- [ ] Hydrogen Storage (Not sure as PYQs not much there)

U4

- Types of Nano materials, Properties all (colour, conductivity etc.,)
- Weight, Number & Viscosity Avg. M.Wt
- Epoxy Resin
- Kevlar
- Carbon Fibre
- All Chemical r/ns
- PMMA
- Factors affecting Polymers (Especially T.S)
- Conducting & Biodegradable Polymers
- OLED Construction & Theory & Working, Applications Adv & Disadv

  (No Types of OLED)
- 4-6 Principles of Green Chemistry